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SUMMARY

AI Engineer (Integrated M.Tech, Sept 2025) building tool-using research agents, hybrid RAG pipelines, and evaluation harnesses with Python, FastAPI, and TypeScript/React. Ships deployed, containerized, test-covered systems (681 automated pytest tests across 4 projects, 3 live on GCP Cloud Run) and measures LLM and retrieval quality with RAGAS and reference-labelled benchmarks. Open-sourced a 12-role multi-agent engineering team for Claude Code: [SoftwareEngineeringAgents](#).

CORE COMPETENCIES & SKILLS

AI & Agentic Systems — Tool-Calling Agents, Multi-Model Verification, Hybrid RAG (Dense + BM25), Reciprocal Rank Fusion (RRF), ChromaDB, Sentence Transformers, LLM APIs (OpenRouter, Groq), Composio, scikit-learn

Evaluation & Reliability — RAGAS, Reference-Labelled Benchmarks, Quote Grounding, Retry-and-Repair, Cost/Budget Guards, pytest (681 Automated Tests), CI/CD (GitHub Actions, ruff, mypy)

Backend & Systems — Python (AsyncIO), FastAPI, Pydantic, REST APIs, Docker, GCP Cloud Run, SQLite (FTS5), PostgreSQL, Git

Full-Stack & Frontend — TypeScript, React 18, Vite, Streamlit

PROJECTS & INITIATIVES

Toolkit Buildability Research — Evidence-Grounded Research Agent *Python, OpenRouter, Composio, Pydantic, pytest*

- Built a tool-using agent that assesses 100 SaaS apps as AI-agent toolkits: allow-listed search/fetch tool chokepoint, schema-bound extraction with up to 2 repair calls, verbatim-quote grounding, and a USD budget cap with a per-call JSONL run log.
- Added a verification pass (cross-family second model, judge on disputed fields only, re-research capped at 5 tool calls). On a 20-app, 140-field blind reference set: field accuracy 0.743 to 0.779, verdict accuracy 14/20 to 17/20, quote grounding across all 100 apps 91.5% to 99.4%; reported the auth-method regression (0.55 to 0.40).
- All runs to date: 567 LLM and 1,220 tool calls for \$4.38; reference labels from an independent agent, hand spot-checked 10/10; 258 automated pytest tests. [\[Live\]](#) [\[GitHub\]](#)

Financial Regulatory Intelligence Engine — Hybrid Dense + Sparse RAG *Python, FastAPI, ChromaDB, BM25, RAGAS, Docker*

- Architected hybrid retrieval over 12,075 chunks from 12 BFSI PDFs (RBI, SEBI, IRDAI circulars and five annual reports), fusing dense MiniLM embeddings with BM25 via Reciprocal Rank Fusion (RRF; $k = 60$).
- Built offline retrieval and RAGAS harnesses (10 labelled questions): MRR 0.678 to 0.703 and RAGAS faithfulness 0.744 to 0.788 (dense to hybrid), with a separate judge model; documented that hybrid lowered context precision (0.470 to 0.287).
- Deployed containerized FastAPI service with model fallback chain, a low-confidence flag, and page-level citations; 82 automated pytest tests with CI. [\[Live\]](#) [\[GitHub\]](#)

GoalOS — AI Executive Coach & Goal Management System *React 18, TypeScript, FastAPI, SQLite FTS5, ChromaDB*

- Engineered a local-first coaching app with keyword-rule intent classification that pre-fetches matching user data for one model call, a registry of 8 tools across 4 domain namespaces (9/9 direct-invocation benchmark, including unregistered-tool rejection), and a persisted session blackboard.
- Implemented 5-factor memory retrieval (SQLite FTS5, ChromaDB cosine, recency half-life decay, importance, access frequency) and deterministic local fallback, with a recorded reason, when offline, rate-limited, or without user consent.
- Shipped with per-call telemetry and CI (ruff, mypy, pytest, Docker build); 280 automated pytest tests. [\[Live\]](#) [\[GitHub\]](#)

Customer Support Analytics — Multi-Task ML & Agentic Triage *Python, FastAPI, scikit-learn, Pydantic, Docker*

- Built a two-tier triage system over 200K tickets: Tier 1 scikit-learn models (82.1% priority accuracy, R-squared = 0.7343 resolution-time regression) handled 69.7% of a 1,000-ticket benchmark without escalation, at about 17–18 ms median local inference.
- Tier 2 LLM escalation fires only for low confidence (< 0.67), predicted resolution > 185 h, or a high-risk Enterprise segment; Pydantic-validated JSON output through a Groq, then OpenRouter, then OpenAI provider chain with deterministic heuristic fallback.
- Deployed FastAPI endpoints (POST /predict_*, POST /triage_agent) on Cloud Run; 61 automated pytest tests with CI. [\[Live\]](#) [\[GitHub\]](#)

EDUCATION

MTech (Integrated) in Software Engineering — Vellore Institute of Technology (VIT), Chennai Jul 2020 – Sep 2025
CGPA: 7.89 / 10.0 | **Certification:** Master Data Science Program (4 months) — HCL GUVI, IITM Incubated (IITM Pravartak program), Jun 2026